

## Warm-up 1

When is your small group office hour?

## Warm-up 2

Given a function  $f : \mathbb{R} \rightarrow \mathbb{R}$ , which of the following condition(s) on  $f$  are necessary for the existence of a function  $g : \mathbb{R} \rightarrow \mathbb{R}$  so that  $g(f(x)) = x$ ?

- a)  $f$  must be continuous
- b)  $f$  must be injective (one-to-one)
- c)  $f$  must be surjective (onto)
- d)  $f$  must be linear

## **During class 1**

Compute:

$$\begin{pmatrix} e & g \\ f & h \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} =$$

## **During class 2**

How many multiplications of field elements does it take to multiply two  $2 \times 2$  matrices? What about for  $3 \times 3$  matrices, or  $n \times n$ ?